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- ROLL NO : S008
- CLASS : SYIT
- SUBJECT : Python for Data Science
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```
import requests
```

- Extract Headings Write a Python program using BeautifulSoup to extract and print all headings from a webpage.

```
from bs4 import BeautifulSoup

# Define the target URL
url = "https://www.bmw.in/en/index.html"

# Send an HTTP request to the webpage
response = requests.get(url)

# Parse the HTML content using BeautifulSoup
soup = BeautifulSoup(response.text, 'html.parser')

# Find all <h2> tags
h2_headings = soup.find_all('h2')

# Extract and print the text from each <h2> heading
for heading in h2_headings:
    print(heading.text.strip())
```

```
BMW 7 Series Sedan
Price starting at ₹1,95,00,000*
Find your BMW.
THE FIRST EVER BMW i5
THE NEW X1.
THE X6 M60i
THE ALL-NEW
BMW Special Offers.
```

- Extract Links Write a Python program using BeautifulSoup to extract all tags and print their link text and URLs.

```
from bs4 import BeautifulSoup

# Define the target URL
url = "https://www.passiton.com/inspirational-quotes"

# Send an HTTP request to the webpage
response = requests.get(url)

# Parse the HTML content using BeautifulSoup
soup = BeautifulSoup(response.text, 'html.parser')

# Find all <a> tags instead of <h2>
a_tags = soup.find_all('a')

# Extract and print the link text and URL from each <a> tag
for tag in a_tags:
    link_text = tag.text.strip()
    link_url = tag.get('href')

    # Sirf wahi links print karein jinka URL aur Text dono valid ho
    if link_text and link_url:
        print(f"Text: {link_text} | URL: {link_url}")
```

```
Text: Pass It On® | The Foundation for a Better Life | URL: /
Text: QUOTES | URL: /inspirational-quotes
Text: VIDEOS | URL: /inspirational-stories-tv-spots
Text: Official Pass It On® Videos | URL: /inspirational-stories-tv-spots
Text: ArtCenter College of Design PSAs | URL: /artcenter-psas
Text: BILLBOARDS | URL: /inspirational-sayings-billboards
Text: STORIES | URL: /positive-good-news-stories
Text: Positive Good News Stories | URL: /positive-good-news-stories
```

Text: NEW Vol. 2 PassItOn® eBook | URL: /free-ebook
 Text: PODCAST | URL: /inspirational-audio-stories
 Text: RADIO | URL: /radio
 Text: BLOG | URL: /passiton-blog
 Text: SCHOOLS | URL: /poster-requests/new
 Text: FREE Posters | URL: /poster-requests/new
 Text: PassItOn® Stories eBook | URL: /free-ebook
 Text: Inspirational Stories | URL: /positive-good-news-stories
 Text: PDF Poster Downloads | URL: /pass-it-on-downloads
 Text: Bookmark Downloads | URL: /pass-it-on-downloads#passiton_bookmarks
 Text: Spanish | Español | URL: <https://www.pasala.org/>
 Text: Portuguese | Português | URL: <https://www.umavidamelhor.org/>
 Text: Chinese | 中文 | URL: <https://www.values.cn.com/>
 Text: Quotes | URL: /inspirational-quotes
 Text: Videos | URL: /inspirational-stories-tv-spots
 Text: Official Videos | URL: /inspirational-stories-tv-spots
 Text: Art Center PSAs | URL: /artcenter-psas
 Text: Billboards | URL: /inspirational-sayings-billboards
 Text: Stories | URL: /positive-good-news-stories
 Text: Good News Stories | URL: /positive-good-news-stories
 Text: FREE eBook | URL: /free-ebook
 Text: Podcast | URL: /inspirational-audio-stories
 Text: Radio | URL: /radio
 Text: Blog | URL: /passiton-blog
 Text: Schools | URL: /poster-requests/new
 Text: FREE Posters | URL: /poster-requests/new
 Text: FREE Downloads | URL: /pass-it-on-downloads
 Text: INTEGRITY | URL: /inspirational-quotes/7654-real-integrity-is-doing-the-right-thing
 Text: APPRECIATION | URL: /inspirational-quotes/7655-integrity-is-choosing-courage-over-comfort
 Text: APPRECIATION | URL: /inspirational-quotes/3983-yesterday-is-history-tomorrow-is-a-mystery
 Text: APPRECIATION | URL: /inspirational-quotes/6639-sometimes-you-will-never-know-the-value-of-a
 Text: APPRECIATION | URL: /inspirational-quotes/3072-the-best-way-to-pay-for-a-lovely-moment-is-to
 Text: APPRECIATION | URL: /inspirational-quotes/4298-appreciation-is-a-wonderful-thing-it-makes
 Text: APPRECIATION | URL: /inspirational-quotes/6794-what-counts-can-t-always-be-counted-what-can
 Text: BELIEVE IN YOURSELF | URL: /inspirational-quotes/8359-believe-in-yourself-and-go-for-it
 Text: BELIEVE IN YOURSELF | URL: /inspirational-quotes/7963-your-self-worth-is-determined-by-you-you-don-t
 Text: BELIEVE IN YOURSELF | URL: /inspirational-quotes/8271-magic-is-believing-in-yourself-if-you-can-do
 Text: BELIEVE IN YOURSELF | URL: /inspirational-quotes/4548-put-all-excuses-aside-and-remember-this-you
 Text: BELIEVE IN YOURSELF | URL: /inspirational-quotes/6966-it-is-our-privilege-and-our-adventure-to
 Text: KINDNESS | URL: /inspirational-quotes/6856-kindness-is-a-language-which-the-deaf-can-hear
 Text: KINDNESS | URL: /inspirational-quotes/8349-how-do-we-change-the-world-one-random-act-of
 Text: KINDNESS | URL: /inspirational-quotes/8165-extend-yourself-in-kindness-to-other-human
 Text: KINDNESS | URL: /inspirational-quotes/6382-ask-yourself-have-you-been-kind-today-make
 Text: KINDNESS | URL: /inspirational-quotes/8368-you-can-be-rich-in-spirit-kindness-love-and
 Text: HONESTY | URL: /inspirational-quotes/8476-honesty-is-often-very-hard-the-truth-is-often
 Text: HONESTY | URL: /inspirational-quotes/8004-don-t-ever-regret-being-honest-period
 Text: HONESTY | URL: /inspirational-quotes/8098-the-time-is-always-right-to-do-what-is-right
 Text: HONESTY | URL: /inspirational-quotes/8475-authenticity-is-a-collection-of-choices-that-we
 Text: HONESTY | URL: /inspirational-quotes/7793-honesty-is-the-fastest-way-to-prevent-a-mistake
 Text: PURPOSE | URL: /inspirational-quotes/7304-the-two-most-important-days-in-your-life-are

Seaborn

```
import seaborn as sns
```

- Bar Chart Create a Seaborn bar chart showing the sales of 5 different products.

```
import matplotlib.pyplot as plt

# Data for 5 different products and their sales
products = ['Laptop', 'Airpods', 'Mobile', 'Oven', 'Smart Watch']
sales = [350, 220, 480, 150, 310]

# Set the background style
sns.set_theme(style="whitegrid")

# Create the bar chart
plt.figure(figsize=(8, 5))
sns.barplot(x=products, y=sales, palette="muted")

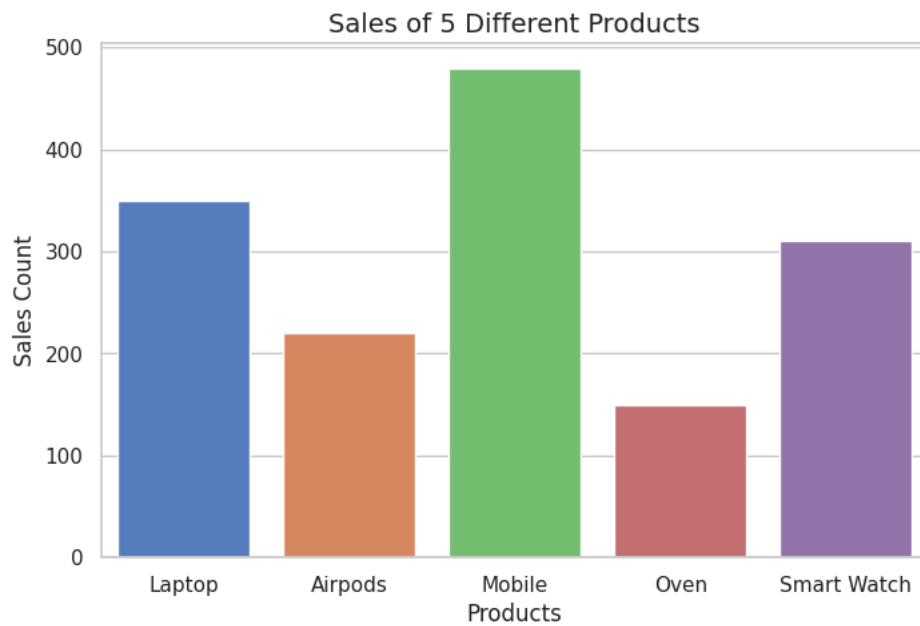
# Add labels and title
plt.xlabel('Products', fontsize=12)
plt.ylabel('Sales Count', fontsize=12)
plt.title('Sales of 5 Different Products', fontsize=14)

# Display the chart
plt.show()
```

```
/tmp/ipykernel_2011/2069233969.py:12: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and

```
sns.barplot(x=products, y=sales, palette="muted")
```



- Scatter Plot Create a Seaborn scatter plot showing the relationship between students' study hours and their marks.

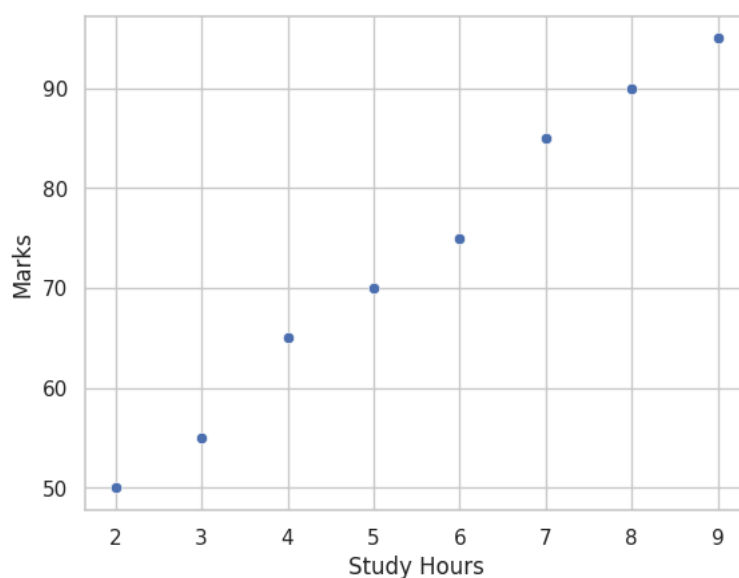
```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
study_hours = [2, 3, 4, 5, 6, 7, 8, 9]
marks = [50, 55, 65, 70, 75, 85, 90, 95]
```

```
sns.scatterplot(x=study_hours, y=marks)
```

```
plt.xlabel('Study Hours')
plt.ylabel('Marks')
```

```
plt.show()
```



Histogram Create a Seaborn histogram to display the distribution of students' marks.

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Students ke marks ka data aur histogram plot karna
```

```
sns.displot([45, 55, 65, 75, 85, 95], kind="hist")
```

```
plt.show()
```

